

Computational Microelectronics

Lecture 5

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L5

Homework#5

- Due: AM08:00, September 21
- Problem#1
 - Calculate the surface potential as a function of V_g .

Φ_m is 4.3 eV.

The p-type doping concentration is $1.5 \times 10^{18} \text{ cm}^{-3}$.

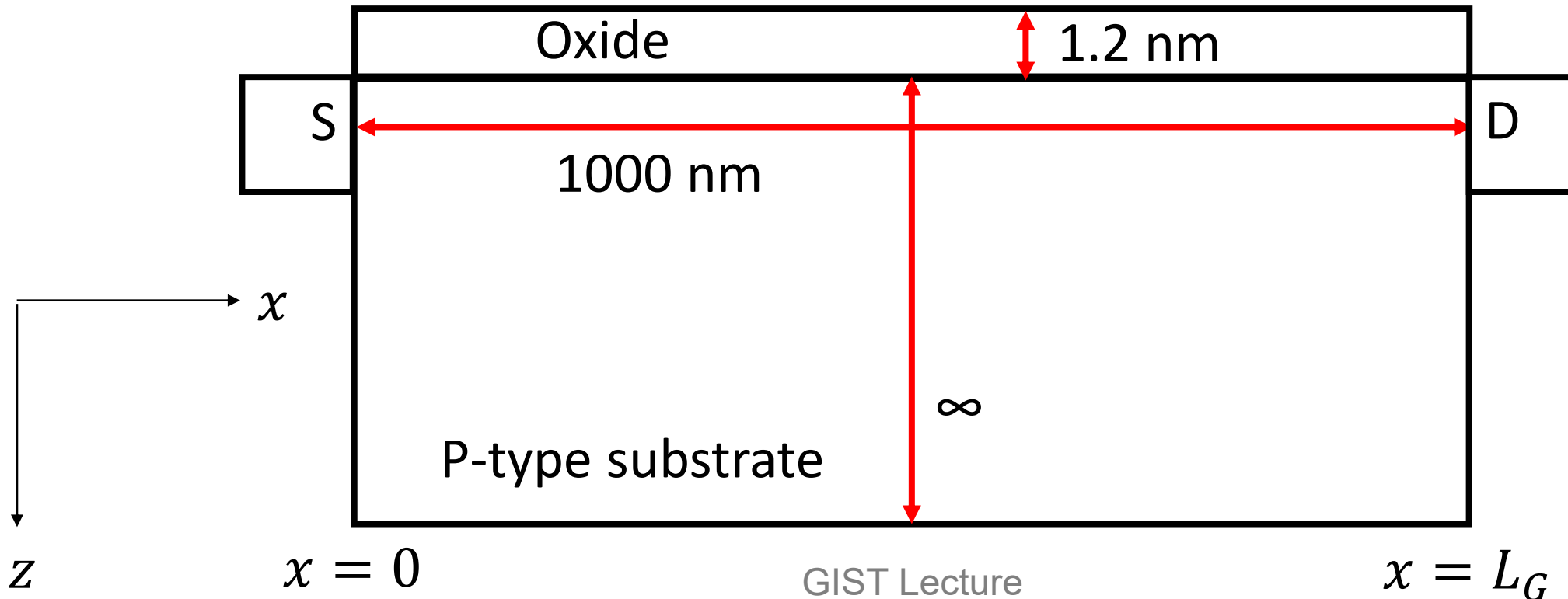
The oxide thickness is 1.2 nm.

Assume 300 K.

Long-channel MOSFET

- Example) $L_G = 1\text{ }\mu\text{m}$ (x is now for the channel direction.)
 - Source/drain contacts set the electron quasi-Fermi potential.

$$\phi_n(0) = V_S \text{ and } \phi_n(L_G) = V_D$$



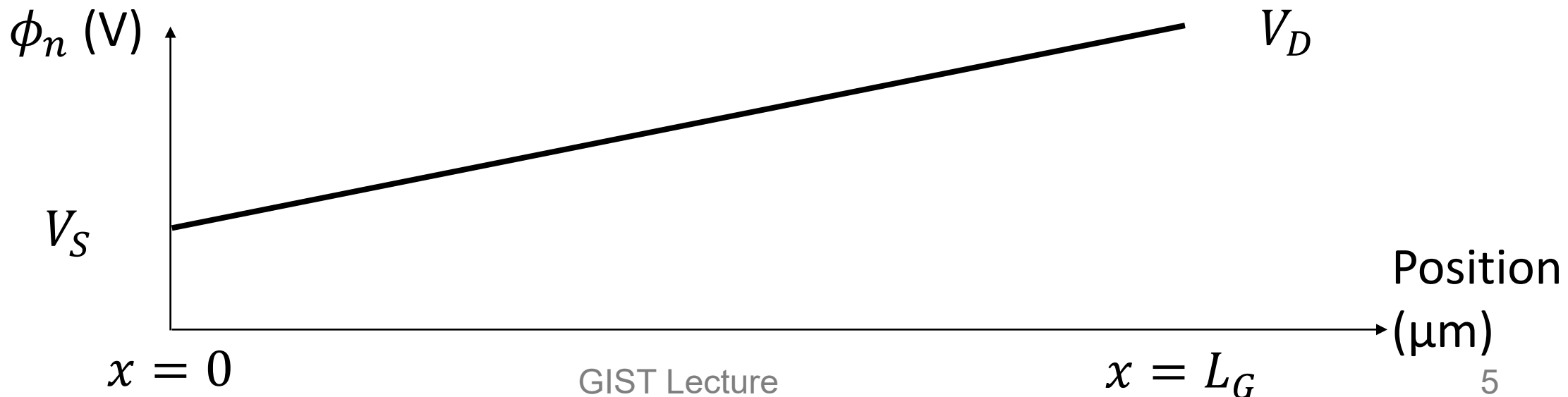
We must calculate $\phi_n(x)$.

- It is obtained by solving the electron continuity equation.

$$\frac{d}{dx} J_{n,x} = \frac{d}{dx} (Q_n v_{n,x}) = 0$$

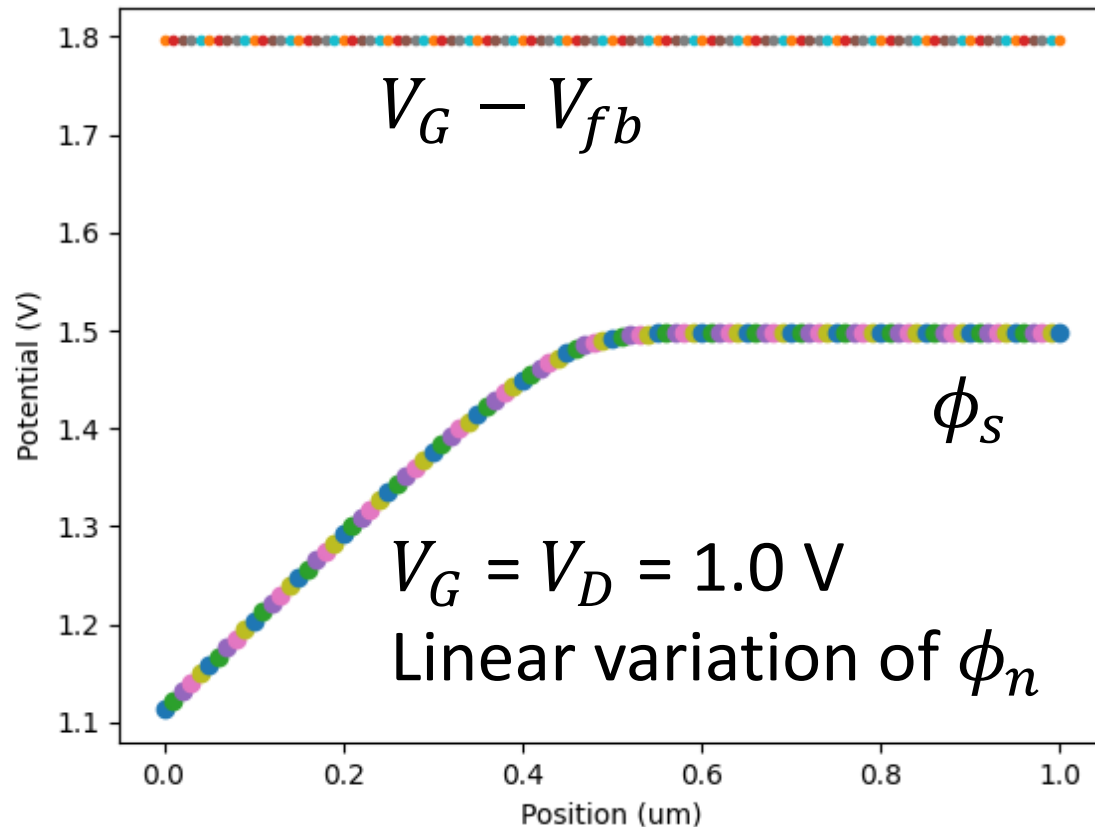
- But, for the moment, assume a linear variation:

$$\phi_n(x) = V_S + (V_D - V_S) \frac{x}{L_G} \text{ and } \frac{d\phi_n}{dx} = \frac{V_D - V_S}{L_G}$$



Under a fixed $\phi_n(x)$,

- It is straightforward to calculate ϕ_s .
 - Your Homework#5 code can solve it! So, we also have Q_s .



Φ_m is 4.3 eV.

The p-type doping concentration is $1.5 \times 10^{18} \text{ cm}^{-3}$.

The oxide thickness is 1.2 nm.

How about Q_n ? (It is not Q_s .)

- We must integrate the electron density.
 - z_i should be sufficiently large.

$$Q_n(x) = -q \int_0^{z_i} n(x, z) dz$$

- There, $\bar{\phi} = \delta \approx 0$.

$$\begin{aligned} Q_n(x) &= -q \int_{\phi_s}^{\delta} \frac{n_{int}^2}{N_a} \exp\left(\frac{q(\bar{\phi} - \phi_n)}{k_B T}\right) \frac{dz}{d\bar{\phi}} d\bar{\phi} \\ &= -q \int_{\delta}^{\phi_s} \frac{\frac{n_{int}^2}{N_a} \exp\left(\frac{q(\bar{\phi} - \phi_n)}{k_B T}\right)}{E_z(\bar{\phi}, \phi_n)} d\bar{\phi} \end{aligned}$$

Can we calculate $E_z(\bar{\phi}, \phi_n)$?

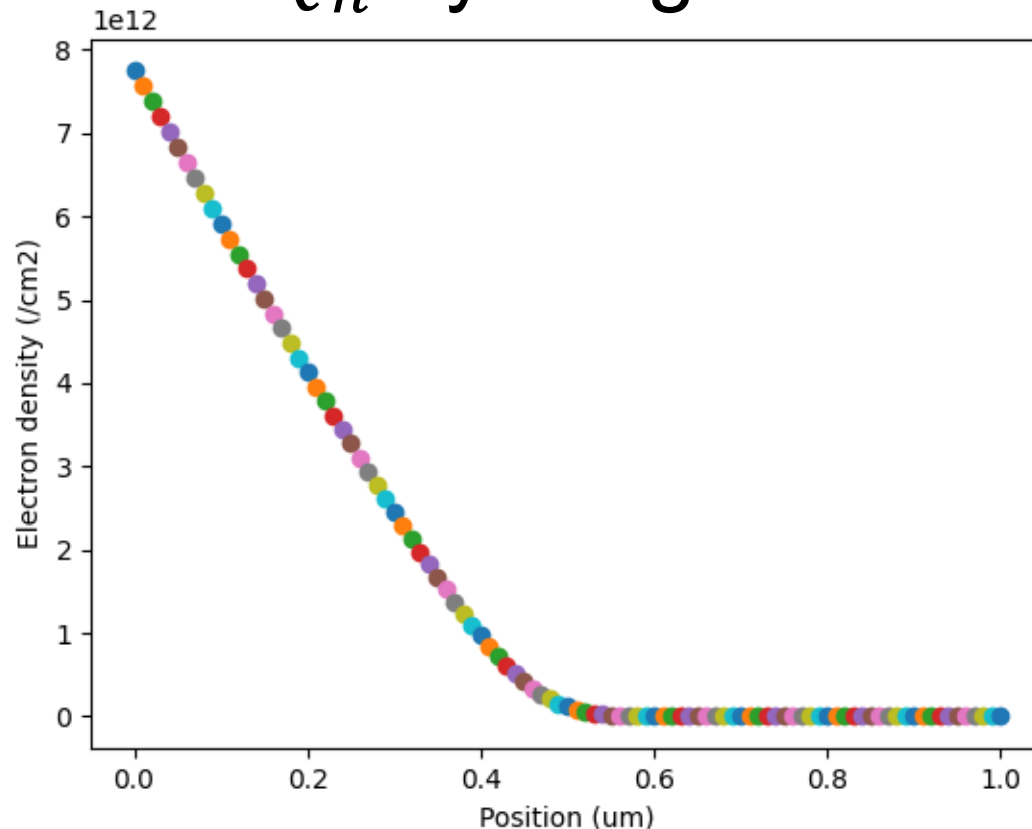
- Yes.

$$E_z(\bar{\phi}, \phi_n)$$

$$= \sqrt{\frac{2k_B T n_{int} / \epsilon_{si}}{\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\bar{\phi}}{k_B T}\right) - 1 + \frac{q\bar{\phi}}{k_B T} \right\} + \exp\left(\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\phi_n}{k_B T}\right) \left(\exp\left(\frac{q\bar{\phi}}{k_B T}\right) - 1 \right) - \frac{q\bar{\phi}}{k_B T} \right\}}}$$

Procedure to calculate Q_n

- 1) Calculate ϕ_s by solving the MOS equation.
 - As we do in Homework#5
- 2) Calculate Q_n by integration



$V_G = V_D = 1.0$ V
Linear variation of ϕ_n

Thank you!